

Alex Sharma

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PROFESSIONAL SUMMARY

Alex Sharma is a backend-leaning software engineer based in Kathmandu, Nepal, with 5 years of professional experience (2019–2024) designing, building, and deploying data-intensive web applications and internal machine learning tooling. Alex is comfortable owning a feature end-to-end — from initial API design through production deployment — and brings a strong engineering culture orientation: thorough documentation, mentoring junior engineers, and raising team-wide code quality.

Core technical identity: backend engineering (Python/FastAPI/Django, Node.js), REST and GraphQL API design, data pipelines, and applied ML/LLM tooling including Retrieval-Augmented Generation (RAG) systems. Secondary strength in cloud infrastructure (AWS), CI/CD automation, and PostgreSQL/Redis data architecture. Personal interests include developer productivity tooling, knowledge management systems, and mentoring.

TECHNICAL SKILLS

Programming Languages

Python (primary, 5 years), JavaScript/TypeScript (4 years), SQL (5 years), Go (basic/learning). Python is used daily for backend services, scripting, data processing, and ML integrations. TypeScript is used for Node.js backend services and React-based admin dashboards.

Backend Frameworks & API Design

FastAPI (current primary framework), Django and Django REST Framework, Node.js with Express. Experienced in RESTful API design principles and GraphQL schema design. Has migrated a production codebase from Flask to FastAPI, resulting in improved type safety, auto-generated docs, and increased test coverage.

Data & Machine Learning

Pandas (data wrangling and ETL), scikit-learn (classical ML models), PyTorch (basic — neural network training and fine-tuning), LangChain (LLM orchestration, chains, and agents), vector databases including ChromaDB (local, open-source) and Pinecone (managed cloud). Has built a production Retrieval-Augmented Generation (RAG) system for internal enterprise knowledge retrieval.

Cloud & Infrastructure

AWS services: EC2 (compute), S3 (object storage), Lambda (serverless functions). Docker (containerisation, multi-stage builds, docker-compose for local dev environments). PostgreSQL (relational database, query optimisation, migrations). Redis (in-memory caching, pub/sub messaging for event-driven architectures). Nginx (reverse proxy configuration for deployed services).

DevOps & CI/CD

GitHub Actions (workflow automation, CI/CD pipeline authoring). Has reduced a team's deployment time from 25 minutes to 6 minutes through pipeline optimisation. Comfortable with infrastructure as code concepts and deployment automation.

Other Skills

Git and GitHub (version control, code review workflows). Agile/Scrum methodology (two-week sprint cycles, sprint planning, retrospectives). Technical writing (runbooks, onboarding guides, architecture decision records). Mentoring junior engineers (pair programming, weekly code review sessions, career development conversations).

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Nimbus Analytics

Kathmandu, Nepal (fully remote) | March 2023 – Present | Industry: Data Analytics / SaaS | Team size: ~20 engineers

Nimbus Analytics is a Kathmandu-based SaaS company building data analytics and business intelligence tooling for enterprise clients across South and Southeast Asia. As a Senior Software Engineer, Alex leads backend feature development, drives API design decisions, maintains ML-powered internal tooling, and mentors junior engineers across the team.

Key Responsibilities and Achievements:

- **Reporting API Redesign:** Led the full redesign of the company's internal reporting API. Identified N+1 query bottlenecks through profiling, introduced query-level caching using Redis with TTL-based invalidation, and added cursor-based pagination to handle large datasets. Result: average API response time dropped from 1.8 seconds to 340 milliseconds — an 81% reduction — directly improving perceived performance for all clients using the reporting dashboard.
- **RAG-Based Internal Knowledge Base:** Architected and built a Retrieval-Augmented Generation (RAG) system for internal use using LangChain as the orchestration layer and ChromaDB as the vector store. Documents (onboarding guides, runbooks, architecture docs) are chunked, embedded, and stored as dense vectors. Queries are answered by retrieving semantically similar chunks and passing them to an LLM for answer synthesis. Outcome: new-hire onboarding time reduced by approximately 30%, with engineers self-serving answers to common questions rather than interrupting senior staff.
- **Junior Engineer Mentorship:** Acted as primary mentor for two junior engineers during their first six months at the company. Conducted weekly 1-on-1 code review sessions, introduced them to the team's coding standards, and guided them through their first independent feature releases. Both engineers progressed to independently owning features within six months.
- **Flask to FastAPI Migration:** Planned and executed the migration of legacy Flask microservices to FastAPI, including rewriting endpoints to use `async/await`, adding Pydantic request/response models for type safety and auto-documentation, and expanding the test suite. Test coverage improved from 45% to 82% over the course of the migration.
- **API Design & Documentation:** Championed a team-wide standard for OpenAPI-compliant documentation using FastAPI's native schema generation. Wrote architecture decision records (ADRs) for major technical choices, building a shared internal knowledge base.

Software Engineer — Bagmati Softworks

Kathmandu, Nepal (on-site) | July 2020 – February 2023 | Industry: Logistics Technology | Team size: ~35 engineers

Bagmati Softworks is a Kathmandu-based software company specialising in logistics and supply-chain technology for clients across Nepal and the broader South Asian market. As a Software Engineer, Alex took on ownership of the core REST API layer for a logistics tracking platform and led key infrastructure improvements including an event-driven notification system and full CI/CD pipeline automation.

Key Responsibilities and Achievements:

- **Logistics Tracking Platform APIs:** Developed, maintained, and extended REST APIs for a logistics tracking platform serving over 15,000 daily active users. Responsibilities included designing new endpoints, writing database migrations, optimising slow PostgreSQL queries, and ensuring backward compatibility for mobile and web clients. Handled on-call incidents for the API layer.
- **Event-Driven Notification System:** Designed and implemented a real-time notification system using Redis pub/sub messaging. When delivery status events occurred (package picked up, in transit, delivered, failed), events were published to Redis channels and consumed by a notification microservice that sent SMS and in-app alerts. Result: missed delivery alerts reduced by 40%, measurably improving end-customer satisfaction scores.
- **Agile Process Participation:** Collaborated closely with product managers and designers within two-week Agile sprint cycles. Contributed to sprint planning, wrote user stories, participated in sprint reviews and retrospectives, and maintained a healthy backlog of technical debt items.
- **CI/CD Pipeline Automation:** Authored and maintained GitHub Actions workflows for automated testing, linting, Docker image builds, and deployment. Reduced total deployment time from 25 minutes to 6 minutes by parallelising test jobs, adding build caching, and eliminating manual deployment steps.
- **Cross-Team Collaboration:** Worked with the mobile team and third-party logistics API providers to design integration contracts, handle webhook ingestion, and ensure reliable data exchange under high-load conditions.

Junior Developer — Bagmati Softworks

Kathmandu, Nepal (on-site) | August 2019 – June 2020 | Industry: Logistics Technology | First professional role

Alex joined Bagmati Softworks immediately after completing a B.E. in Computer Engineering at Tribhuvan University. The junior role focused on building internal tooling, fixing bugs across the main Django codebase, and writing unit tests to improve platform reliability.

Key Responsibilities and Achievements:

- **Internal Admin Dashboards:** Built internal admin dashboards using React (frontend) and Django REST Framework (backend API). Dashboards were used by operations staff to manage shipments, view delivery metrics, and manually trigger notifications.
- **Bug Fixing and Testing:** Triaged and resolved bugs across the company's main Django monolith. Wrote unit and integration tests using pytest and Django's TestCase framework, contributing to the overall improvement of code quality and regression prevention.
- **Onboarding and Learning:** Completed the company's onboarding process, learned Agile workflows, and adapted to a professional engineering team environment within the first month of joining.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Engineering

Tribhuvan University, Institute of Engineering, Kathmandu, Nepal | 2015 – 2019 | Graduated with Distinction

Four-year undergraduate degree in Computer Engineering at Nepal's largest and most established university. Graduated with Distinction, reflecting strong academic performance across all four years.

Relevant coursework included: Data Structures and Algorithms, Database Systems (relational and NoSQL), Operating Systems, Artificial Intelligence and Machine Learning fundamentals, Software Engineering principles (SDLC, design patterns, testing), Computer Networks, and Discrete Mathematics.

PERSONAL PROJECTS

Course Recommender Chatbot

Personal / Side Project | 2022–2023

Built a conversational course recommendation chatbot using the OpenAI API for natural language understanding and Gradio for the web-based chat interface. The system ingests a scraped dataset of online course listings (titles, descriptions, provider, topics, ratings) and recommends relevant courses based on a user's free-text description of their interests or learning goals. Demonstrates practical experience with LLM API integration, prompt engineering, and building lightweight ML-powered user-facing tools.

Technologies used: Python, OpenAI API (GPT-3.5/4), Gradio, Pandas, web scraping (BeautifulSoup).

Expense Tracker REST API

Personal / Side Project | 2023

Designed and built a production-deployed personal finance tracking backend API. The system allows users to log expenses, categorise transactions, set monthly budgets, and view spending summaries via a REST API. Deployed on an AWS EC2 instance behind an Nginx reverse proxy with SSL termination. Database is PostgreSQL with Alembic for schema migrations.

Technologies used: Python, FastAPI, PostgreSQL, Alembic, AWS EC2, Nginx, Docker. Demonstrates ability to take a backend project from local development through to cloud deployment.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **AWS Certified Cloud Practitioner (2022)** — Issued by Amazon Web Services. Validates foundational understanding of AWS cloud services including EC2, S3, Lambda, IAM, VPC, and cloud cost models. Demonstrates commitment to cloud infrastructure knowledge beyond on-the-job experience.
- **Deep Learning Specialization** — DeepLearning.AI / Coursera (2021) — Five-course specialization taught by Andrew Ng. Covered neural network fundamentals, training deep networks, convolutional neural networks (CNNs), sequence models (RNNs, LSTMs), and practical ML project structuring. Provides the foundational deep learning knowledge underpinning PyTorch and ML tooling work in professional projects.

LANGUAGES

English — Full professional proficiency (fluent). Used daily for technical documentation, code reviews, cross-team written communication, and reading technical literature. **Nepali** — Native speaker. **Hindi** — Conversational proficiency. Useful for communication with clients and partners in India and the broader South Asian region.

WORKING STYLE AND APPROACH

Alex Sharma approaches engineering with a documentation-first mindset: systems, decisions, and runbooks are written down so that knowledge is shared across the team rather than siloed. This manifests in writing architecture decision records, maintaining onboarding guides, and contributing to internal wikis.

Alex places high value on code quality and review culture. In both individual contributor and informal tech-lead roles, Alex has run regular code review sessions, enforced linting and formatting standards via CI, and introduced testing requirements that meaningfully raised coverage. Alex is also experienced in giving and receiving constructive feedback in a respectful, growth-oriented way.

As a mentor, Alex prioritises building the mentee's confidence and independence rather than providing answers directly. Techniques include pair programming, Socratic questioning during code reviews, and setting structured short-term goals for skill development.

Alex is comfortable working remotely (fully remote since March 2023) and is experienced with asynchronous communication norms, written status updates, and maintaining high output without in-person oversight.